

Eye-Gaze Text Entry Experimental Analysis

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Automated report covering typing performance, gaze smoothing, dwell-to-click behavior, head pose, method comparison and statistical analysis.

Methodology & Equations

Experimental Setup

This experiment is designed to analyze and compare text entry performance, particularly between Traditional Keyboard and Eye-Gaze Keyboard modalities. Both techniques use the same predefined phrases to ensure experimental control and reproducibility.

For each trial, the following variables are ideally recorded: Target phrase, Entered phrase, Completion time, Number of errors, Accuracy, Words per minute (WPM), and Characters per second (CPS).

Typing Performance Equations

WPM: Calculated as the Total Typed Characters divided by 5, all divided by the Duration in Minutes.

Accuracy (%): Calculated as $(1 - (\text{Uncorrected Errors} / \text{Total Characters Typed})) * 100$.

1. Dataset Overview

Item	Value
CSV files	1
Total rows/trials	4
Columns	15
Participants	Not detected
Methods	1
Trials	4

Input files

experiment_overt_brittick_20260907_162721.csv

Detected Variables

Metric	CSV Column
method	method
trial	trial_number
characters_typed	char_count

2. Typing Performance

No standard typing metrics were detected.

WPM follows the supplied experimental definition: (total typed characters / 5) divided by duration in minutes. Accuracy follows: $(1 - \text{uncorrected errors} / \text{total characters typed}) \times 100$.

4. Gaze Smoothing and Dwell-to-Click Algorithm

Exponential Moving Average (EMA) Smoothing

To eliminate micro-tremors in raw gaze coordinates, the system applies Exponential Moving Average smoothing. The formula used is:

$$S(t) = \alpha * Raw(t) + (1 - \alpha) * S(t-1)$$

Where the smoothing factor $\alpha = 0.12$. A small α weights the smoothed history heavily ($1 - 0.12 = 0.88$), which suppresses sudden jitter while still allowing the estimate to gradually track genuine gaze movement.

Raw and/or smoothed gaze X/Y columns were not detected, so gaze jitter could not be calculated from this dataset.

5. Head Pose Analysis

Yaw/pitch columns were not detected.

6. Dwell-to-Click / Selection Analysis

Selection/dwell columns were not detected. If your CSV contains these values, rename them to a common form such as dwell_time, selection_total, selection_success or false_clicks.

7. Correlation Analysis

Not enough compatible numeric variables were available for correlation analysis.

11. Automated Findings

- The dataset did not contain enough recognized variables to produce automated comparative findings.

Interpretation Notes

These findings are descriptive unless a statistical test confirms significance. The report does not assume that a difference is meaningful merely because its mean is larger or smaller. For research reporting, inspect sample size, variance, participant matching, task order, and p-values before making strong conclusions.

12. Data Quality and Limitations

Missing-data percentage for detected variables:

Variable	Missing
method	0.0%
trial	0.0%
characters_typed	0.0%

Column names are detected heuristically. Review the 'Detected Variables' table above to ensure the mapping is correct. Derived errors are calculated as simple position-wise character mismatches when explicit error counts are unavailable; this is not equivalent to every possible text-entry error metric.